

Bayesian Learning

Lecture 5 - Linear Regression. Regularization.

Mattias Villani

Department of Statistics
Stockholm University



mattiasvillani.com



@matvil



@matvil



mattiasvillani

Lecture overview

- Linear regression
- Regularization priors

Linear regression

- The linear regression model in **matrix form**

$$\underset{(n \times 1)}{\mathbf{y}} = \underset{(n \times k)(k \times 1)}{\mathbf{X}\beta} + \underset{(n \times 1)}{\varepsilon}$$

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$
$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = \begin{pmatrix} x_{11} & \cdots & x_{1k} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nk} \end{pmatrix}$$

- Usually $x_{i1} = 1$, for all i . β_1 is the intercept.
- **Likelihood**

$$\mathbf{y} | \beta, \sigma^2, \mathbf{X} \sim N(\mathbf{X}\beta, \sigma^2 I_n)$$

Linear regression - uniform prior

- Standard **non-informative prior**: uniform on $(\beta, \log \sigma^2)$

$$p(\beta, \sigma^2) \propto \sigma^{-2}$$

- **Joint posterior** of β and σ^2 :

$$\begin{aligned}\beta | \sigma^2, \mathbf{y} &\sim N(\hat{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}) \\ \sigma^2 | \mathbf{y} &\sim \text{Inv-}\chi^2(n - k, s^2)\end{aligned}$$

where $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ and $s^2 = \frac{1}{n-k} (\mathbf{y} - \mathbf{X}\hat{\beta})^\top (\mathbf{y} - \mathbf{X}\hat{\beta})$.

- **Simulate** from the joint posterior by simulating from

- ▶ $p(\sigma^2 | \mathbf{y})$
- ▶ $p(\beta | \sigma^2, \mathbf{y})$

- **Marginal posterior** of β :

$$\beta | \mathbf{y} \sim t_{n-k}(\hat{\beta}, s^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

Linear regression - conjugate prior

■ Joint prior for β and σ^2

$$\begin{aligned}\beta|\sigma^2 &\sim N(\mu_0, \sigma^2\Omega_0^{-1}) \\ \sigma^2 &\sim \text{Inv} - \chi^2(\nu_0, \sigma_0^2)\end{aligned}$$

■ Posterior

$$\begin{aligned}\beta|\sigma^2, \mathbf{y} &\sim N(\mu_n, \sigma^2\Omega_n^{-1}) \\ \sigma^2|\mathbf{y} &\sim \text{Inv} - \chi^2(\nu_n, \sigma_n^2)\end{aligned}$$

$$\mu_n = \left(\mathbf{X}^\top \mathbf{X} + \Omega_0\right)^{-1} \left(\mathbf{X}^\top \mathbf{X} \hat{\beta} + \Omega_0 \mu_0\right)$$

$$\Omega_n = \mathbf{X}^\top \mathbf{X} + \Omega_0$$

$$\nu_n = \nu_0 + n$$

$$\sigma_n^2 = \left(\nu_0 \sigma_0^2 + \mathbf{y}^\top \mathbf{y} + \mu_0^\top \Omega_0 \mu_0 - \mu_n^\top \Omega_n \mu_n\right) / \nu_n$$

Bike share data

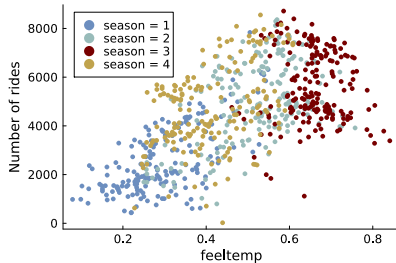
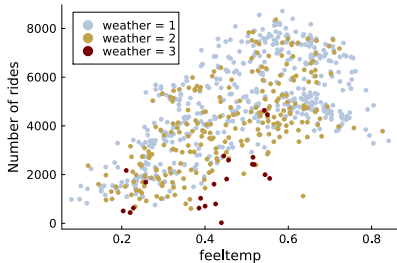
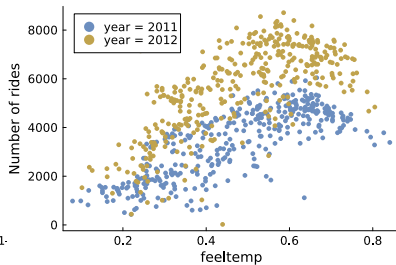
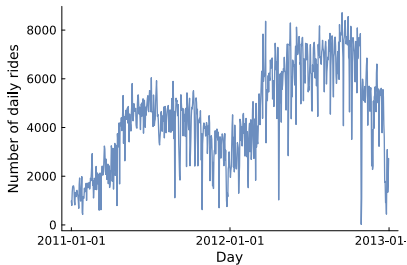
- **Bike share data.** Predict the number of bike rides.
- Response variable: number of rides on 731 days.

variable	description	type	values	comment
nrides	# of rides	counts	$\{0, 1, \dots\}$	min= 22, max= 8714
feeltemp	perceived temp	cont.	$[0, 1]$	min= 0.07, max= 0.85
hum	humidity	cont.	$[0, 1]$	min= 0.00, max= 0.98
wind	wind speed	cont.	$[0, 1]$	min= 0.02, max= 0.51
year	year	binary	$\{0, 1\}$	year 2011 = 0
season	season	cat.	$\{1, 2, 3, 4\}$	winter $\rightarrow$ fall
weather	weather	ordinal	$\{1, 2, 3\}$	clear $\rightarrow$ rain/snow
weekday	day of week	cat.	$\{0, \dots, 6\}$	sunday $\rightarrow$ saturday
holiday	holiday	binary	$\{0, 1\}$	holiday = 1

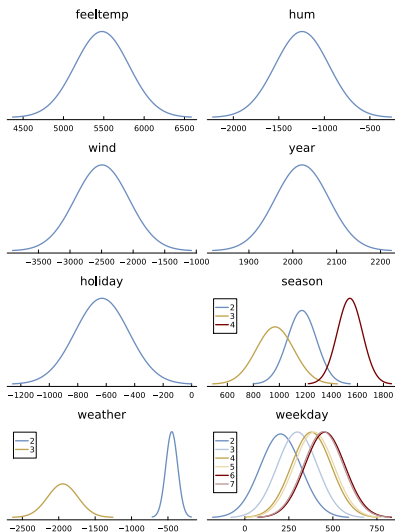
- Prior:

- ▶ $\mu_0 = (1000, 0, \dots, 0)^\top$
- ▶ $\Omega_0 = \frac{\kappa_0}{n} \mathbf{X}^\top \mathbf{X}$ with $\kappa_0 = 1$ (unit information prior)
- ▶ $\sigma_0^2 = 1000^2$ and $\nu_0 = 5$.

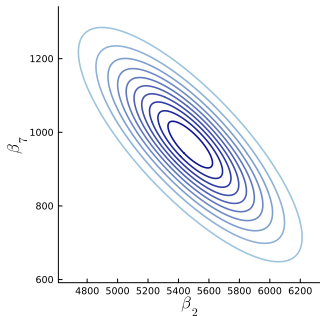
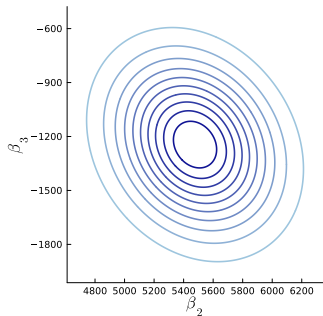
Bike share data



Bike share data - marginal posteriors of β



Bike share data - joint posteriors of β



Interactive - Bayesian regression

prior type: ☒ Zellner g-prior ☐ Identity

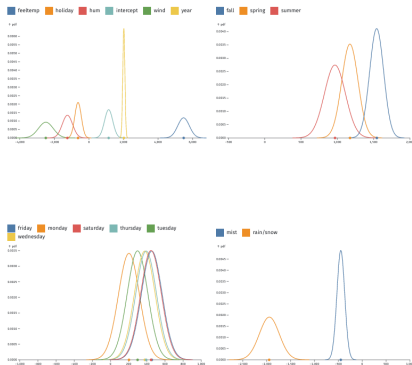
prior sample size, n_0

prior intercept, μ_0

ν_0

σ_0^2

Marginal posteriors for selected variables. Least squared estimates are indicated by points.



Mattias Villani Bayesian linear regression- bike share data

Observable

Polynomial regression

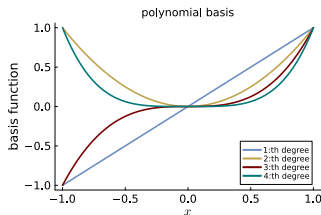
Polynomial regression

$$f(x_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_k x_i^k, \quad \text{for } i = 1, \dots, n.$$

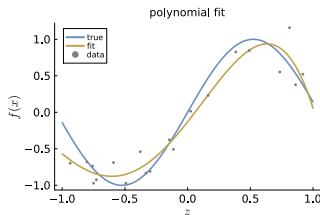
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$$\mathbf{x}_i = (1, x_i, x_i^2, \dots, x_i^k)^\top$$

Still **linear in $\boldsymbol{\beta}$** and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$. Bayes unchanged.



1.00	-1.00	1.00	-1.00	1.00
1.00	-0.90	0.81	-0.73	0.66
1.00	-0.80	0.64	-0.51	0.41
1.00	-0.70	0.49	-0.34	0.24
1.00	-0.60	0.36	-0.22	0.13
1.00	-0.50	0.25	-0.12	0.06
1.00	-0.40	0.16	-0.06	0.03
1.00	-0.30	0.09	-0.03	0.01
1.00	-0.20	0.04	-0.01	0.00
1.00	-0.10	0.01	-0.00	0.00
1.00	0.00	0.00	0.00	0.00
1.00	0.10	0.01	0.00	0.00
1.00	0.20	0.04	0.01	0.00
1.00	0.30	0.09	0.03	0.01
1.00	0.40	0.16	0.06	0.03
1.00	0.50	0.25	0.12	0.06
1.00	0.60	0.36	0.22	0.13
1.00	0.70	0.49	0.34	0.24
1.00	0.80	0.64	0.51	0.41
1.00	0.90	0.81	0.73	0.66
1.00	1.00	1.00	1.00	1.00



Polynomials are **global** basis functions. **Local** basis preferred.

Predictive distribution regression

Predictive density conjugate Gaussian linear regression

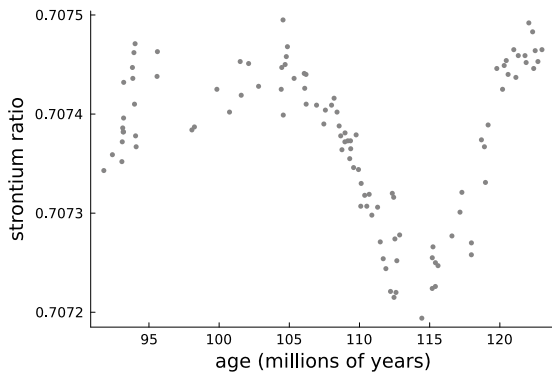
Model: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$

Posterior: $\boldsymbol{\beta} | \sigma^2, \mathbf{y}, \mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_n, \sigma^2 \boldsymbol{\Omega}_n^{-1})$
 $\sigma^2 | \mathbf{y}, \mathbf{X} \sim \text{Inv-}\chi^2(\nu_n, \sigma_n^2)$

Predictive density for $\tilde{n}$ observations with covariate matrix $\tilde{\mathbf{X}}$:
 $\tilde{\mathbf{y}} | \tilde{\mathbf{X}}, \mathbf{y}, \mathbf{X} \sim t\left(\tilde{\mathbf{X}}\boldsymbol{\mu}_n, \sigma_n^2(\mathbf{I}_{\tilde{n}} + \tilde{\mathbf{X}}\boldsymbol{\Omega}_n^{-1}\tilde{\mathbf{X}}^\top), \nu_n\right)$

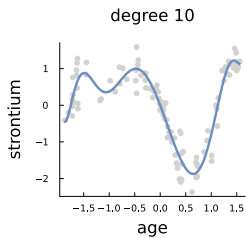
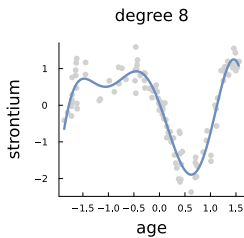
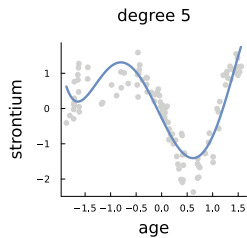
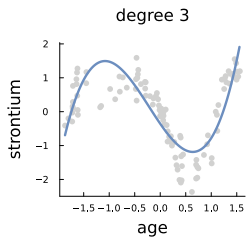
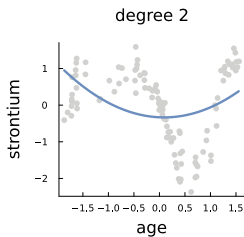
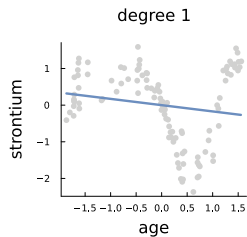
where the posterior hyperparameters $\boldsymbol{\mu}_n$, $\boldsymbol{\Omega}_n$, σ_n^2 and ν_n defined in Box 5.3.

Fossil data

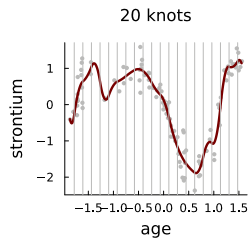
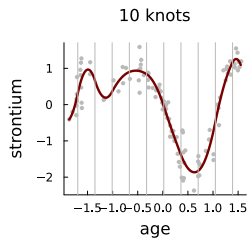
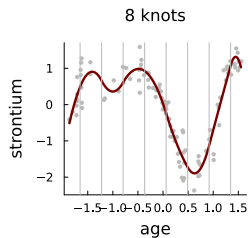
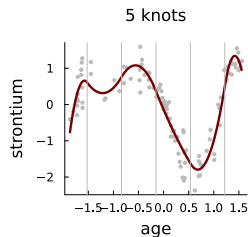
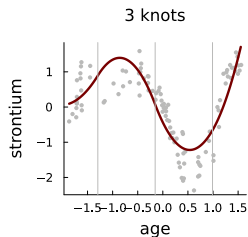
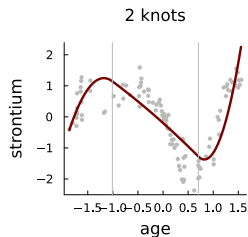


From Ruppert, Wand and Carroll (2003). Semiparametric regression.

Polynomial regression - fossil data



Quadratic spline - fossil data



Regularization prior - Ridge

- Splines: too many knots leads to **over-fitting**.
- **Shrinkage/regularization prior**

$$\beta_i | \sigma^2 \stackrel{\text{iid}}{\sim} N\left(0, \frac{\sigma^2}{\lambda}\right)$$

- Larger λ gives smoother fit. Note: $\Omega_0 = \lambda I$ in conjugate prior.

Ridge regression uses an independent Gaussian prior

The prior

$$\beta_j | \sigma^2 \stackrel{\text{iid}}{\sim} N(0, \sigma^2 / \lambda) \quad (12.6)$$

$$\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2) \quad (12.7)$$

for the linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 I_n) \quad (12.8)$$

implies a posterior mean for $\boldsymbol{\beta}$ equal to the ridge estimator

$$\hat{\boldsymbol{\beta}}_{L_2} = (\mathbf{X}^\top \mathbf{X} + \lambda I_p)^{-1} \mathbf{X}^\top \mathbf{y}. \quad (12.9)$$

Ridge shrinks toward zero

- Ridge regression estimator

$$\tilde{\beta} = (\mathbf{X}^T \mathbf{X} + \lambda I_p)^{-1} \mathbf{X}^T \mathbf{y}$$

- Shrinkage toward zero

$$\text{As } \lambda \rightarrow \infty, \tilde{\beta} \rightarrow 0$$

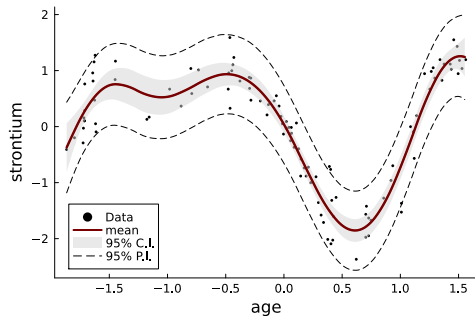
- When $\mathbf{X}^T \mathbf{X} = I_p$

$$\tilde{\beta} = \frac{1}{1 + \lambda} \hat{\beta} = (1 - \phi) \hat{\beta}$$

- Shrinkage factor

$$\phi = \frac{\lambda}{1 + \lambda}$$

Spline with Gaussian prior - fossil data



Regularization prior - Lasso

- **Lasso** is equivalent to posterior mode under Laplace prior

$$\beta_i | \sigma^2 \stackrel{\text{iid}}{\sim} \text{Laplace} \left(0, \frac{\sigma^2}{2\lambda} \right)$$

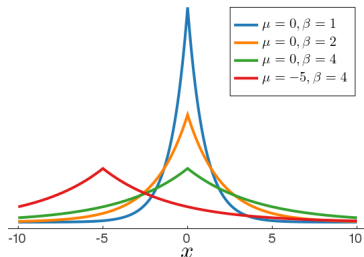
Laplace distribution

$X \sim \text{Laplace}(\mu, \beta)$ for $X \in \mathbb{R}$.

$$p(x) = \frac{1}{2\beta} \exp \left(-\frac{|x - \mu|}{\beta} \right)$$

$$\mathbb{E}(X) = \mu$$

$$\mathbb{V}(X) = 2\beta^2$$



- The **Bayesian shrinkage** prior is **interpretable**. **Not ad hoc**.
- Laplace distribution have heavy tails.
- **Laplace prior**: many β_i close to zero, but some β_i very large.
- Normal distribution have light tails.
- **Normal prior**: all β_i 's are similar in magnitude.

Learning the shrinkage

- **Cross-validation** used to determine degree of smoothness, λ .
- Bayesian: λ is **unknown** $\Rightarrow$ **use a prior** for λ !
- $\lambda^{-1} \sim \text{Inv-}\chi^2(\omega_0, \psi_0^2)$.
- **Hierarchical** setup:

$$\mathbf{y}|\boldsymbol{\beta}, \sigma^2, \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 I_n)$$

$$\boldsymbol{\beta}|\sigma^2, \lambda \sim N(0, \sigma^2 \lambda^{-1} I_m)$$

$$\sigma^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

$$\lambda^{-1} \sim \text{Inv-}\chi^2(\omega_0, \psi_0^2)$$

so $\boldsymbol{\Omega}_0 = \lambda I_m$.

Regression with learned shrinkage

- The **joint posterior** of β , σ^2 and λ is

$$\beta|\sigma^2, \lambda, \mathbf{y} \sim N(\mu_n, \Omega_n^{-1})$$

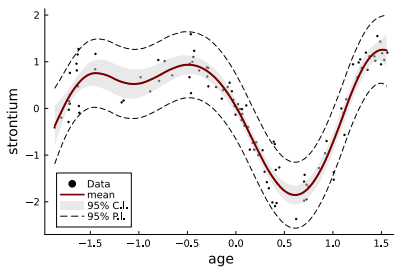
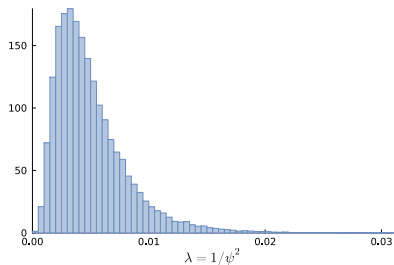
$$\sigma^2|\lambda, \mathbf{y} \sim \text{Inv} - \chi^2(\nu_n, \sigma_n^2)$$

$$p(\lambda|\mathbf{y}) \propto \sqrt{\frac{|\Omega_0|}{|\mathbf{X}^\top \mathbf{X} + \Omega_0|}} \left(\frac{\nu_n \sigma_n^2}{2} \right)^{-\nu_n/2} \cdot p(\lambda)$$

where $\Omega_0 = \lambda I_m$, and $p(\lambda)$ is the prior for λ .

- Or simulate from $p(\beta, \sigma^2, \lambda|\mathbf{y}, \mathbf{X})$ using Gibbs sampling (L7).

Spline with Gaussian prior - fossil data



Horseshoe prior

- Normal and Laplace - only one global shrinkage parameter λ .
- **Global-Local shrinkage**: global + **local shrinkage** for each β_j .
- **Horseshoe prior**:

$$\beta_j | \lambda_j^2, \tau^2 \sim N(0, \tau^2 \lambda_j^2)$$

$$\lambda_j \sim C^+(0, 1)$$

$$\tau \sim C^+(0, 1)$$

- When $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$, posterior mean for β satisfies approximately

$$\mu_{n,j} \approx (1 - \phi_j) \hat{\beta}_j, \text{ where } \phi_j = \frac{1}{1 + (n/\sigma^2) \tau^2 \lambda_j^2}$$

- Implied **prior on shrinkage**

